Commit or Recover

Where and why frozen TransFuser++ fails in CARLA, and whether failures can be predicted

CMPE249 team project - CARLA 0.9.15, Bench2Drive routes - analysis date 2026-11-06

1. Executive summary

- **What ran:** 260 route runs (220 at seed 100 covering all suites, plus the 20 selected focus routes repeated at seeds 200 and 300). 181,390 control steps at 20 Hz, 27.8 km driven, 2.5 h of simulated driving.
- **How often it fails:** 91 of 260 runs (35%) had a failure. Mean driving score 81.7, mean route completion 96.4%.
- **What kind:** 72 vehicle collisions, 4 static, 4 pedestrian, 4 off-route, 7 stuck until timeout. Vehicle-vs-vehicle conflicts are 79% of all failures.
- **Where:** 36 failures happen in a **junction turn / crossing** and 23 in a **lane change / side-swipe**; together 65% of failures. By family, lane merges fail most (52%), then junctions (48%).
- **Is it consistent?** Mostly. Of the 20 focus routes run at three seeds, 8 failed every time and 9 never failed; only 3 were mixed. A same-seed rerun matched on 17 of 19. Failures are driven by the scenario, not by random chance.
- **Can we predict them?** Only weakly from TF++'s own outputs. Step-level AUROC looks good (0.86) but most of that is "the car is moving"; on moving steps only it drops to 0.75, versus 0.70 for speed alone. At a usable 1 false alarm per minute the best model warns at least 1 s ahead for 24% of failures, versus 21% for speed alone.
- **Recommendation:** the monitor needs scene information. TF++ already predicts bounding boxes of other actors (detect_boxes=1); feeding those (distance, closing speed, time-to-collision to the nearest predicted box) to the GRU is the most promising next step. Planner-uncertainty signals only rise in the last 1-2 s.

2. What ran

Setup

- Model: frozen TransFuser++ (carla_garage, all_towns checkpoint model_0030_0). No fine-tuning.
- Simulator: CARLA 0.9.15 with AdditionalMaps, run headless on SJSU coe-hpc3 A100 nodes via Slurm (2 nodes x 4 workers, shared claim queue so routes spread across workers).
- Routes and scenarios: Bench2Drive route set, evaluated with the Bench2Drive leaderboard and scenario runner (the carla_garage leaderboard silently skips several Bench2Drive scenario types, so its early results were discarded).
- Logging: per-step JSONL from the agent (controls, speed, planner outputs, uncertainty signals) plus a privileged ground-truth block (nearby actors, junction state, lane geometry) used only for labeling and situation analysis, never as a monitor input. Infraction events are recorded with their simulator frame.
- Robustness: a 15-minute stall watchdog was added to the runner because CARLA occasionally crashes mid-route and the evaluator would otherwise wait forever; hung routes are killed and retried.

Runs

Run	Contents	Seed	Routes	Failure rate	Driving score	Route completion
all_seed100	All suites (route families + other Bench2Drive routes)	100	200	32%	83.0	96.7%
selected_seed100	20 selected focus routes	100	20	45%	79.1	95.3%
selected_seed200	20 selected focus routes	200	20	50%	75.7	95.3%
selected_seed300	20 selected focus routes	300	20	45%	77.7	95.2%
Total			260	35%	81.7	96.4%

Each route+seed pair is counted once; where a route was in both the all-suites run and the selected run at seed 100 the duplicate was dropped. The 20 selected focus routes cover junction, lane-merge and obstacle families, two to three routes per scenario type.

All infraction events recorded

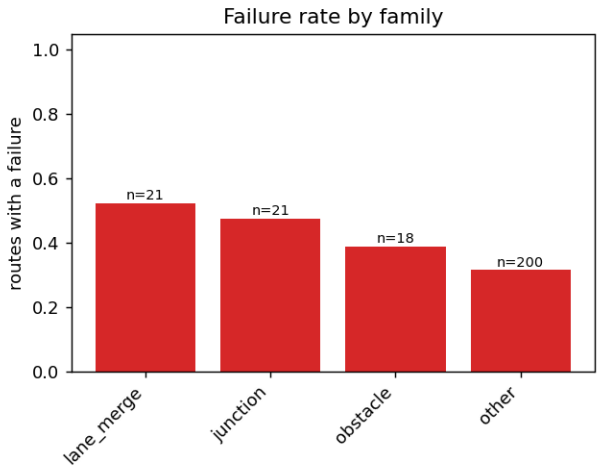
Event type	Events	Routes affected	Counted as failure?
MIN_SPEED_INFRACTION	4800	218	no (minor)
COLLISION_VEHICLE	88	55	yes
COLLISION_STATIC	12	5	yes
OUTSIDE_ROUTE_LANES_INFRACTION	9	8	yes
TRAFFIC_LIGHT_INFRACTION	6	6	no (minor)
YIELD_TO_EMERGENCY_VEHICLE	5	5	no (minor)
COLLISION_PEDESTRIAN	4	4	yes
VEHICLE_BLOCKED	1	1	yes

A run is a failure if it has a collision, leaves the route lanes, is blocked, or times out stuck. Minimum-speed, red-light and emergency-vehicle infractions lower the driving score but are not safety failures. The route's first failure is the one analysed below.

3. Where TF++ fails

By scenario family

Family	Routes	Failure rate	Driving score	Route completion
lane_merge	21	52.4%	77.6	100.0%
junction	21	47.6%	75.3	92.9%
obstacle	18	38.9%	80.0	92.6%
other	200	31.5%	83.0	96.7%

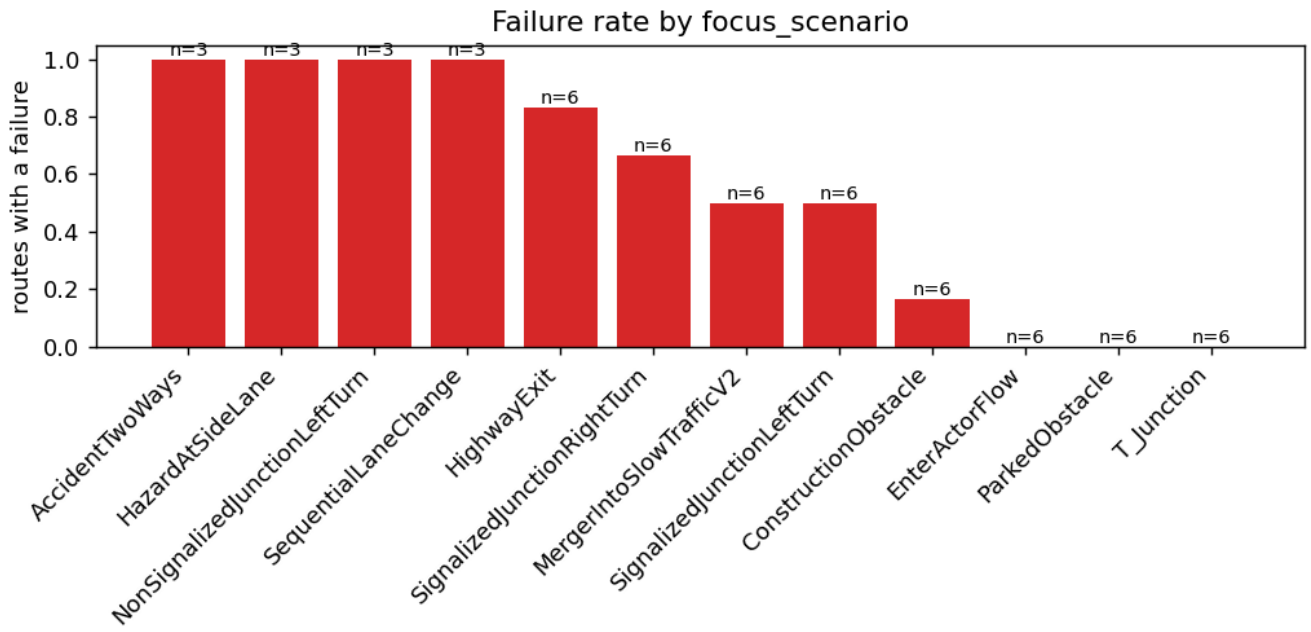


Failure rate by scenario family.

By focus scenario (selected routes, all three seeds)

Focus scenario	Runs	Failure rate	Driving score	Route completion
AccidentTwoWays	3	100%	33.3	55.4%
HazardAtSideLane	3	100%	60.0	100.0%
NonSignalizedJunctionLeftTurn	3	100%	30.1	50.2%
SequentialLaneChange	3	100%	52.0	100.0%
HighwayExit	6	83%	65.5	100.0%
SignalizedJunctionRightTurn	6	67%	73.3	100.0%
MergerIntoSlowTrafficV2	6	50%	80.0	100.0%
SignalizedJunctionLeftTurn	6	50%	80.0	100.0%
ConstructionObstacle	6	17%	93.3	100.0%
EnterActorFlow	6	0%	100.0	100.0%
ParkedObstacle	6	0%	100.0	100.0%
T_Junction	6	0%	95.0	100.0%

AccidentTwoWays, HazardAtSideLane, NonSignalizedJunctionLeftTurn and SequentialLaneChange failed on every run. EnterActorFlow, ParkedObstacle and T_Junction never caused a failure. Unprotected left turns and going around an obstacle into oncoming traffic are the clearest weaknesses.

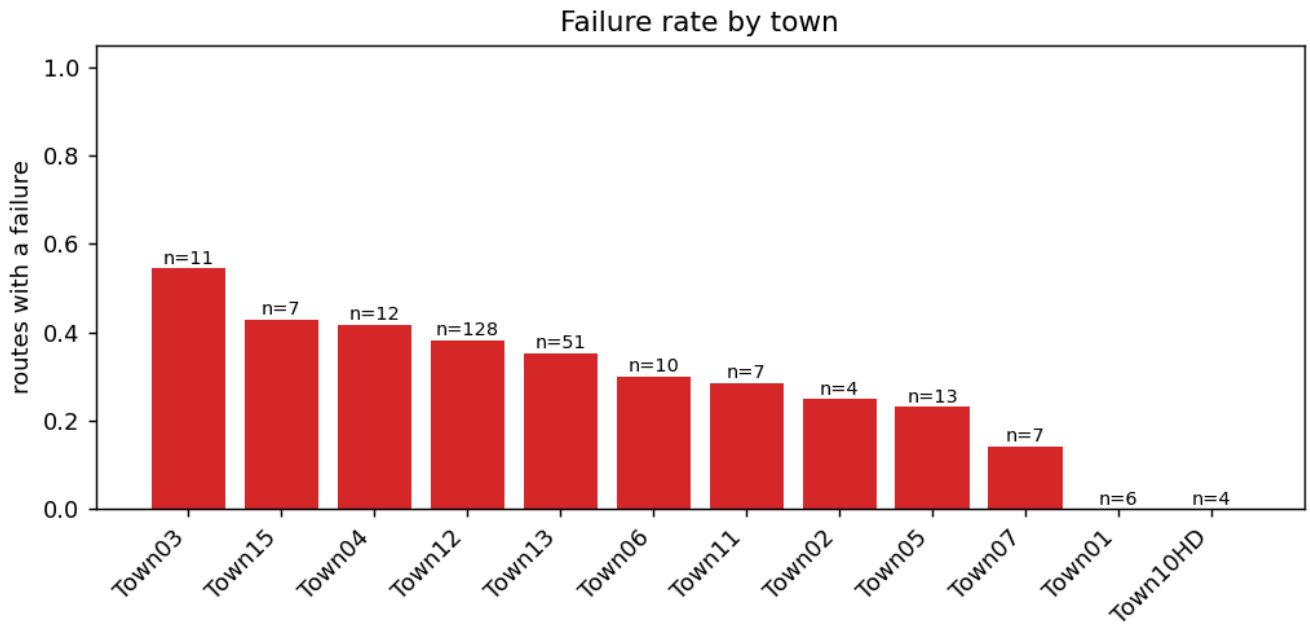


Failure rate by focus scenario.

By town

Town	Routes	Failure rate	Driving score	Route completion
Town03	11	55%	60.9	77.1%
Town15	7	43%	70.5	96.1%
Town04	12	42%	77.6	98.9%
Town12	128	38%	81.7	96.4%
Town13	51	35%	80.9	97.6%
Town06	10	30%	85.6	100.0%
Town11	7	29%	85.0	100.0%
Town02	4	25%	87.5	100.0%
Town05	13	23%	90.7	99.0%
Town07	7	14%	87.2	94.0%
Town01	6	0%	95.0	100.0%
Town10HD	4	0%	100.0	100.0%

Most routes are in Town12 and Town13 (the large Bench2Drive towns), so the town numbers for the small towns rest on 4-13 routes each and should not be over-read. Town03 stands out (55%, completion 77%).



Failure rate by town.

Weather and time of day

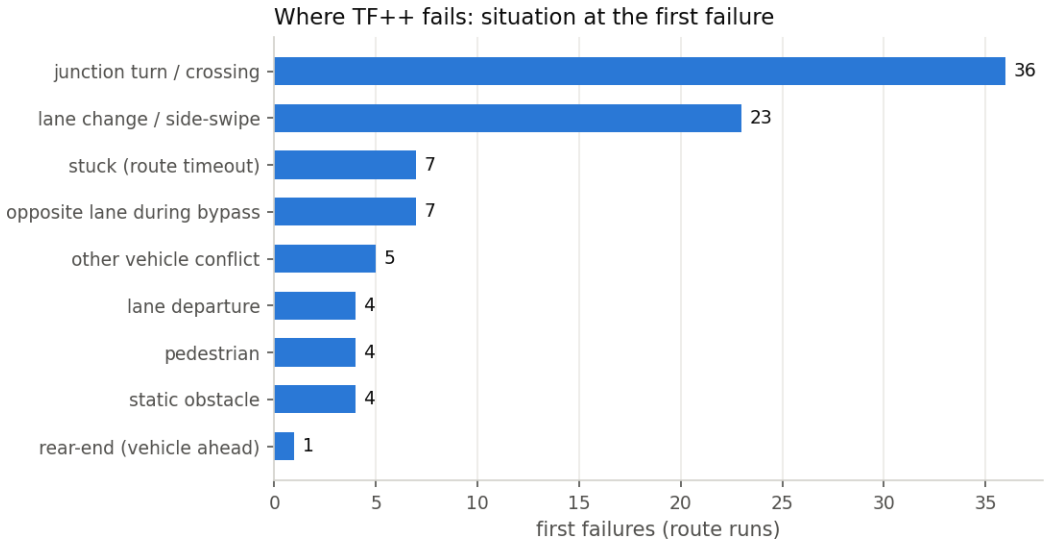
Condition	Routes	Failure rate
Dry (precipitation <= 20)	133	37%
Wet (precipitation > 20)	127	33%
Day (sun above horizon)	201	36%
Night (sun below horizon)	59	31%

Neither rain nor darkness raises the failure rate in this data; the scenario itself matters far more than the weather.

4. In which situations failures happen

Each failure was assigned to one situation with fixed rules over the privileged ground truth at the failure moment (failure type, junction state, heading relative to the lane, current command, nearest actor ahead). Rules are applied in order: stuck, pedestrian, static, lane departure, junction, opposite lane, lane change, rear-end, other.

Situation	Count	Share	Speed at failure (m/s)	Speed 1.5 s before	Braking before	Top scenario types
junction turn / crossing	36	40%	10.5	8.1	19%	MergerIntoSlowTrafficV2, SignalizedJunctionLeftTurn, NonSignalizedJunctionLeftTurn
lane change / side-swipe	23	25%	8.7	9.1	32%	SequentialLaneChange, HighwayExit, HazardAtSideLane
opposite lane during bypass	7	8%	12.2	7.9	15%	AccidentTwoWays, ConstructionObstacleTwoWays
stuck (route timeout)	7	8%	—	—	—	HazardAtSideLane, ParkingExit, StaticCutIn
other vehicle conflict	5	5%	10.4	14.1	43%	VehicleOpensDoorTwoWays, HardBreakRoute, ConstructionObstacle
static obstacle	4	4%	11.8	5.7	7%	ConstructionObstacleTwoWays, InvadingTurn
pedestrian	4	4%	6.6	8.8	45%	DynamicObjectCrossing, ParkingCrossingPedestrian, PedestrianCrossing
lane departure	4	4%	9.2	9.9	31%	VanillaSignalizedTurnEncounterGreenLight, VanillaNonSignalizedTurn, InvadingTurn
rear-end (vehicle ahead)	1	1%	15.7	9.6	0%	MergerIntoSlowTraffic



Failures by situation.

What each situation looks like

- **junction turn / crossing** (36): Collision while inside or just after a junction (left/right turns, crossing flows).
- **lane change / side-swipe** (23): Collision while changing lanes or angled 8-60 deg to the lane (merges, exits).
- **opposite lane during bypass** (7): Collision while in the opposite lane going around a blockage (heading >120 deg).
- **stuck (route timeout)** (7): Never collided, but stopped and never moved on until the route timed out.

- **other vehicle conflict** (5): Vehicle collisions on a straight lane not covered by the rules above (door opening, hard brake).
- **lane departure** (4): Left the route lanes (off-route infraction) on turns.
- **pedestrian** (4): Struck a pedestrian (crossing, parked-car emergence).
- **static obstacle** (4): Struck a static object (construction cones, invading-turn obstacles).
- **rear-end (vehicle ahead)** (1): Hit a vehicle ahead closer than 15 m on a straight lane.

Trends

- **Failures happen at speed and without braking.** In junctions TF++ is at 10.5 m/s when it hits, and braked in only 19% of the 1.5 s before. In other words, it does not see the conflict coming; it commits to the turn.
- **Junction failures accelerate into the collision** (from 8.1 to 10.5 m/s), typical of an unprotected left turn into crossing traffic or merging into a flow it misjudged.
- **Lane-change failures are side-swipes at a 19.3 deg angle** with about 5 actors within 20 m: TF++ cuts in without a gap (SequentialLaneChange, HighwayExit, HazardAtSideLane).
- Bypass failures (AccidentTwoWays, ConstructionObstacleTwoWays) happen with the car fully in the opposite lane (heading ~172 deg off the route lane) meeting oncoming traffic.
- Pedestrian and "other vehicle" cases are the only ones with substantial braking beforehand (45% and 43%): TF++ reacts, but too late.
- Only 30% of collisions occur with the ego strictly inside a junction polygon; the rest of the "junction" cases are just after exiting while the turn completes.

5. Repeatability across seeds

The 20 selected routes were run at seeds 100, 200 and 300 (traffic and scenario randomness change with the seed). A failure that repeats on every seed is a systematic weakness of the policy, not bad luck.

Family	Focus scenario	Route	Seed 100	Seed 200	Seed 300	Failed seeds
junction	NonSignalizedJunctionLeftTurn	2084	FAIL	FAIL	FAIL	3/3
junction	SignalizedJunctionLeftTurn	3936	FAIL	FAIL	FAIL	3/3
junction	SignalizedJunctionLeftTurn	4183	ok	ok	ok	0/3
junction	SignalizedJunctionRightTurn	2050	FAIL	FAIL	FAIL	3/3
junction	SignalizedJunctionRightTurn	26956	ok	FAIL	ok	1/3
junction	T_Junction	26458	ok	ok	ok	0/3
junction	T_Junction	28035	ok	ok	ok	0/3
lane_merge	EnterActorFlow	11715	ok	ok	ok	0/3
lane_merge	EnterActorFlow	27506	ok	ok	ok	0/3
lane_merge	HighwayExit	23658	FAIL	FAIL	FAIL	3/3
lane_merge	HighwayExit	23670	FAIL	ok	FAIL	2/3
lane_merge	MergerIntoSlowTrafficV2	23771	ok	ok	ok	0/3
lane_merge	MergerIntoSlowTrafficV2	24092	FAIL	FAIL	FAIL	3/3
lane_merge	SequentialLaneChange	17563	FAIL	FAIL	FAIL	3/3
obstacle	AccidentTwoWays	1852	FAIL	FAIL	FAIL	3/3
obstacle	ConstructionObstacle	24367	ok	ok	ok	0/3
obstacle	ConstructionObstacle	2509	ok	FAIL	ok	1/3
obstacle	HazardAtSideLane	1790	FAIL	FAIL	FAIL	3/3
obstacle	ParkedObstacle	1773	ok	ok	ok	0/3
obstacle	ParkedObstacle	24757	ok	ok	ok	0/3

- 8 routes failed on all three seeds, 9 never failed, and only 3 (23670, 26956, 2509) were mixed.
- Re-running the same seed gave the same outcome on 17 of 19 routes; the two differences (1790, 26956) show some simulator non-determinism even at a fixed seed.
- For the monitor this is useful: failure-prone situations are reproducible, so training data can be collected deliberately from these routes.

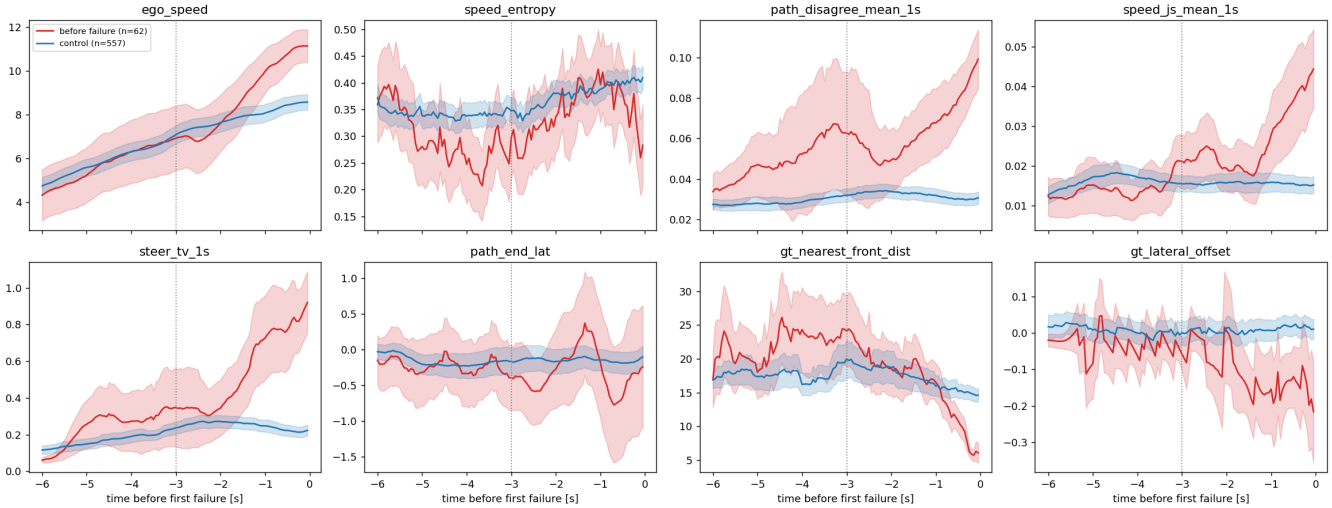
6. Precursors: what changes before a failure

Every logged signal was scored on its own for separating steps in the 3 s before a failure from all other steps, using only steps where the car is moving (>2 m/s), so that "the car is stopped, so it cannot crash" does not inflate the score. AUROC 0.5 is chance.

Signal	Group	Direction	AUROC (moving)	AUPRC
path_end_lat_std_1s	temporal	higher before failure	0.696	0.177
path_curvature	single	higher before failure	0.678	0.174
path_end_lat_std_2s	temporal	higher before failure	0.677	0.160
path_disagree_mean_1s	temporal	higher before failure	0.669	0.167
path_disagree_mean	temporal	higher before failure	0.666	0.162
speed_js_mean_2s	temporal	higher before failure	0.666	0.150
target_speed_scalar	single	higher before failure	0.664	0.152
path_disagree_max_1s	temporal	higher before failure	0.663	0.156
steer_tv_1s	temporal	higher before failure	0.647	0.180
ego_speed	single	higher before failure	0.646	0.148
path_disagree_mean_2s	temporal	higher before failure	0.642	0.159
path_disagree_max_2s	temporal	higher before failure	0.640	0.153
speed_entropy_mean_2s	temporal	lower before failure	0.639	0.132
steer_tv_2s	temporal	higher before failure	0.628	0.172
speed_js_mean_1s	temporal	higher before failure	0.628	0.147

Base rate (share of moving steps within 3 s of a failure): 8.6%.

- **No single signal reaches 0.70.** The best is the jitter of the predicted path end point over the last second (path_end_lat_std_1s, 0.70), followed by path curvature and disagreement between consecutive planned paths.
- Speed and target speed alone score about 0.65-0.66: failures happen when TF++ is driving faster and wants to go faster.
- Speed-head uncertainty (entropy, top probability, margin) is weak (0.59-0.64), and lower entropy, not higher, comes before failures: TF++ is confidently wrong.
- Control-level signals (throttle, brake, steer, jerk, reversals) are close to chance.



Signal traces aligned on the failure moment (t = 0), failures versus matched non-failure windows.



Precursor traces split by situation.

The traces show the key problem for a monitor: planner instability rises only in the last 1-2 s before impact. Up to then, the failing runs look like normal driving.

7. Can we predict failures?

Classifiers were trained to flag steps within 3 s of a failure, using only signals TF++ itself produces. Evaluation is 5-fold cross-validation grouped by route, so a route at another seed never appears in both train and test. 84 collision/off-route failures are scored (stuck timeouts have no impact moment). Four feature sets: single-frame (current outputs), temporal (1-2 s windowed statistics), both, and a speed-only baseline.

Ranking quality

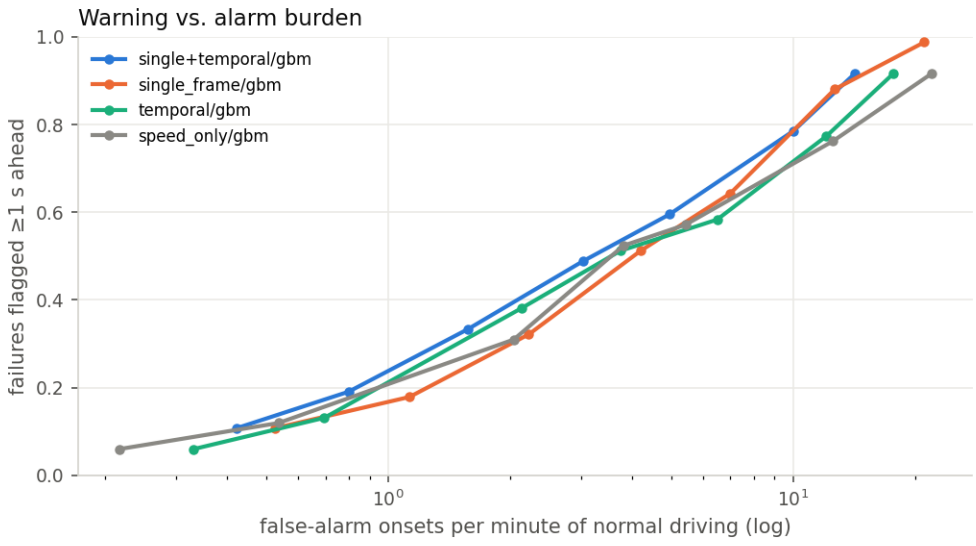
Features	Model	AUROC (all steps)	AUROC (moving only)	AUPRC
single_frame	logreg	0.863	0.750	0.206
single_frame	gbm	0.861	0.749	0.223
temporal	logreg	0.805	0.683	0.139
temporal	gbm	0.841	0.733	0.183
single+temporal	logreg	0.860	0.748	0.198
single+temporal	gbm	0.854	0.740	0.213
speed_only	logreg	0.819	0.661	0.144
speed_only	gbm	0.828	0.701	0.186

All-steps AUROC of 0.85 looks strong, but speed alone gets 0.83. On moving steps the best model is 0.75 against 0.70 for speed alone: the planner and uncertainty signals add only about 0.04-0.05.

Usefulness as an alarm

A monitor is judged by how many failures it warns about early enough to act, at a false-alarm rate a driver could tolerate. A "5% per-step false positive rate" sounds low but works out to about 10 separate false alarms per minute of driving. The table interpolates each GBM's operating curve at 1, 3 and 10 false-alarm onsets per minute.

Features (GBM)	Warned >=1 s @ 1/min	@ 3/min	@ 10/min	Warned >=2 s @ 1/min	@ 3/min	@ 10/min
Single-frame + temporal	24%	49%	79%	15%	28%	54%
Single-frame only	17%	41%	79%	13%	28%	58%
Temporal only	21%	46%	72%	11%	26%	59%
Speed only (baseline)	21%	44%	71%	13%	30%	58%

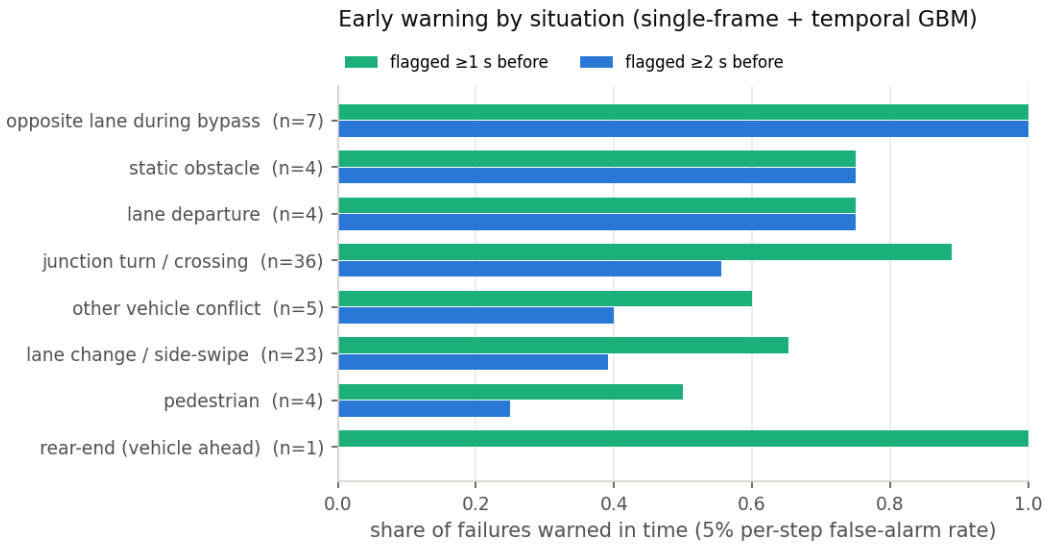


Share of failures warned at least 1 s ahead, against false-alarm onsets per minute.

Which situations can be warned about (combined GBM at the 5% per-step setting)

Situation	Failures	Warned >=1 s	Warned >=2 s	Median lead time
junction turn / crossing	36	89%	56%	2.45 s
lane change / side-swipe	23	65%	39%	1.15 s
opposite lane during bypass	7	100%	100%	3.15 s
other vehicle conflict	5	60%	40%	1.05 s
lane departure	4	75%	75%	4.55 s
pedestrian	4	50%	25%	1.05 s
static obstacle	4	75%	75%	2.43 s
rear-end (vehicle ahead)	1	100%	0%	1.40 s

This setting costs about 10 false alarms per minute, so these are upper bounds. Bypasses into the opposite lane are easy to flag (the car is visibly in the wrong lane for several seconds). Lane changes and pedestrians are the hardest: median lead time near 1 s, too late to recover.



Share of failures warned at least 1 s and 2 s ahead, by situation.

8. What this means for the monitor

- **The failures are predictable in kind but not in timing from TF++'s outputs alone.** We know where TF++ fails (unprotected turns, lane changes into traffic, bypasses into oncoming lanes), and it fails there repeatably, but its own control, planning and uncertainty outputs only change in the last 1-2 s.
- **TF++ is confidently wrong.** Speed-head entropy is lower, not higher, before failures, so uncertainty-based "commit or recover" triggers will miss most failures.
- **Add scene inputs.** TF++ already decodes bounding boxes of surrounding actors (detect_boxes=1 in config.py, decoded in sensor_agent). Distance, closing speed and time-to-collision to the nearest predicted box in the planned path are the natural next features. The privileged ground truth logged in this study can be used to check how good those predicted boxes are.
- **Add context inputs.** Junction approach, the current route command (turn / lane change) and whether the ego is in the opposite lane are cheap to compute and line up with the failure situations.
- **Evaluate honestly.** Report results on moving steps, against a speed-only baseline, and as warnings per failure at a fixed number of false alarms per minute. Step-level AUROC overstates usefulness.
- **Target data collection.** The 8 routes that fail on every seed (2084, 3936, 2050, 23658, 24092, 17563, 1852, 1790) are a ready-made hard set for training and testing the monitor.

9. Caveats

- 84 scored failures is a small sample; per-situation numbers rest on 1-36 cases and confidence intervals are wide.
- The situation labels come from simple rules, not manual review of video. Borderline cases (for example a lane change just after a junction) go to whichever rule fires first.
- All prediction models are offline classifiers on logged signals, not the final GRU monitor, and no intervention was tested in closed loop.
- Route 23670 crashed CARLA repeatedly; its all-suites seed-100 run never completed and is covered by the selected-routes runs instead.
- Some simulator non-determinism remains even at a fixed seed (2 of 19 routes changed outcome on rerun).

10. Files and how to reproduce

What	Where (under ~/CMPE249/Commit-Or-Recover)
Raw run results	carla_garage/results/<suite>_seed<N>/
Route runner (with stall watchdog)	carla_garage/tools/run_bench2drive_routes.py
Dataset builder	analysis/build_dataset.py
Failure analysis	analysis/analyze_failures.py
Deep dive (situations, alarms, seeds)	analysis/deep_dive.py
This PDF	analysis/make_report_pdf.py
Tables, parquet, figures	analysis/data/combined/ (figures in figures/)

Reproduce: `python analysis/build_dataset.py`, then `python analysis/analyze_failures.py --data analysis/data/combined`, `python analysis/deep_dive.py --data analysis/data/combined --repeat carla_garage/results/selected_seed100 carla_garage/results/all_seed100`, and `python analysis/make_report_pdf.py`.